UALBANY

Electrocardiogram Signal–Based Cardiac Abnormality Detection

Dataset

MIT-BIH Arrhythmia Database

The dataset contains 48 ECG recordings from different patients.

Each recording is approximately 30 minutes long and sampled at 360 Hz

Models

Classical ML Model

- Principal Component Analysis (PCA) + Support Vector Machine (SVM)
- Hand-crafted electrocardiogram (ECG) features
- Dimensionality reduction for improved feature representation

Neural Network Model

- Regularized One-Dimensional Convolutional Neural Network (1D CNN)
- Operates directly on raw ECG signals
- Incorporates Batch Normalization and Dropout to improve generalization

Model Architecture

Raw ECG Signal	→	Beat Windowing	→	Standard Scaler	→	PCA	→	SVM	→	Normal / Abnormal
WFDB format .dat / .hea / .atr		180 samples per beat		Normalize amplitude		180 → 30 features		RBF kernel C=10, balanced		Binary classification

PCA — Principal Component Analysis

Purpose: Reduce 180 raw ECG time-steps → 30 key components

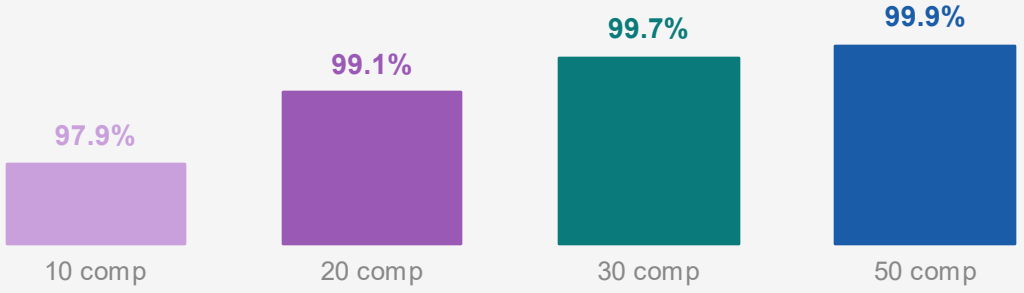
Variance kept: 99.7% (almost no information lost)

How: Finds patterns that vary most across all 112,579 beats

Component 1 → R-peak height (main heartbeat spike)

Component 2 → QRS complex shape

Component 3+ → P-wave, T-wave patterns



Variance Retained vs. Components

SVM — Support Vector Machine

Purpose: Draw the optimal boundary between Normal & Abnormal beats

Kernel: RBF (Radial Basis Function) — handles non-linear boundaries

Key parameters:

Initial SVM	Improved SVM
`C=1.0` `No class weight`	`C=10.0` `class_weight='balanced'`
Abnormal Recall: **76%**	Abnormal Recall: **94%**

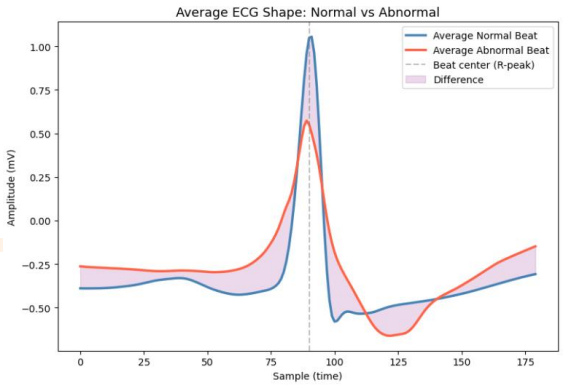
Why it worked: Dataset has 4× more normal beats (90k) than abnormal (22k). Balancing forced the model to treat both classes equally.

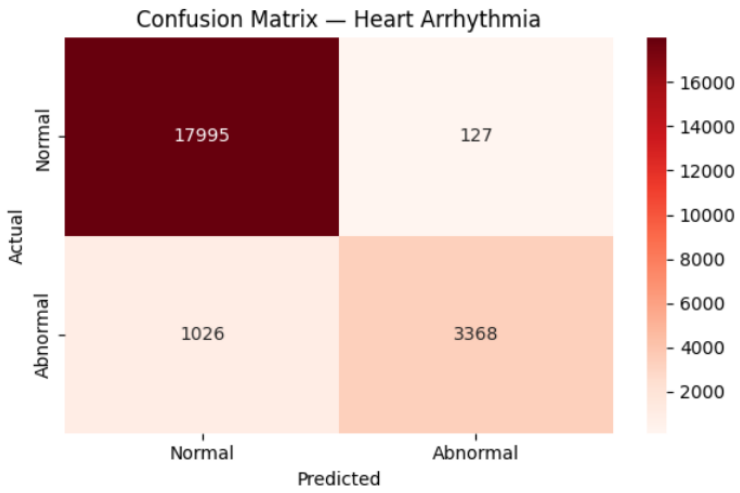
Model Results

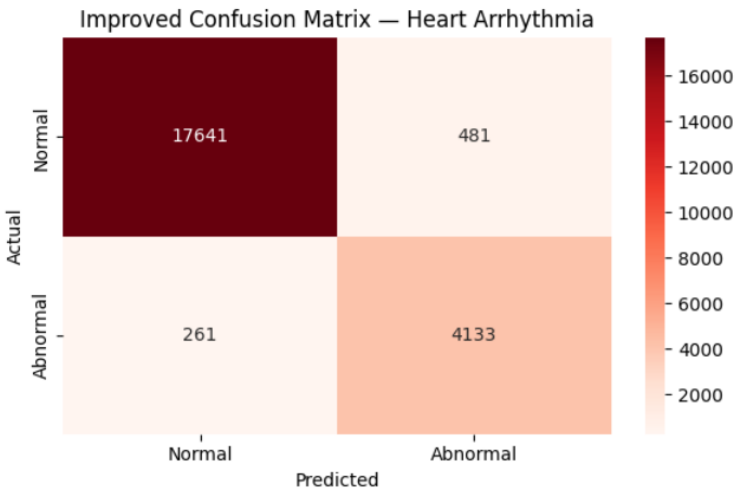
97%	94%	89%	92%
Overall Accuracy	Abnormal Recall	Abnormal Precision	F1-Score

Before vs. After Class Balancing

Metric	Initial Model	Improved Model
Accuracy	95%	97% ↑
Abnormal Recall	76%	94% ↑
Missed Beats	1,050	253 ↓







Key Findings

- Class balancing reduced missed patients from 1,050 → 253 doing this
- PCA retained 99.7% variance using only 30 components Where 10 components changed the variance to 97.9% and 50 components changed the variance to 99.9%
- SVM with RBF kernel C=10 proved most effective rather than increasing the components or samples in SVM as the other 2 options would slow the speed of model
- A defining component that stands out is the average amplitude of the normal heartbeats vs abnormal heartbeats it is seen abnormal beats have an average of .6mV and normal heartbeats averaging at 1mV.
- *Next step: 1D CNN for comparison*

Why 1D CNN for ECG Arrhythmia Detection?

1D Temporal Structure

ECG is a 1-D time series. Conv1D kernels slide along the time axis, directly matching how P, QRS, and T-wave features are sequentially embedded in the signal.

Automatic Feature Extraction

No hand-crafted features needed. The network learns R-peak shape, beat morphology, and rhythm patterns end-to-end from raw windowed samples.

Efficient & Lightweight

Only ~445K parameters across 3 conv blocks. Each beat is a 200-sample window — fast enough for real-time clinical inference.

Handles Class Imbalance

Weighted cross-entropy (1:8) penalises missed arrhythmias more heavily, so the model learns rare patterns that a PCA+SVM pipeline often loses.

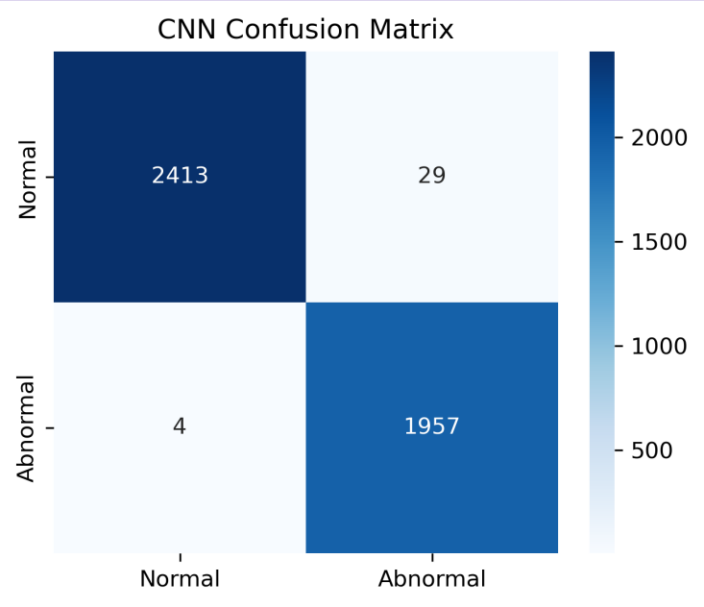
1D CNN Model Architecture

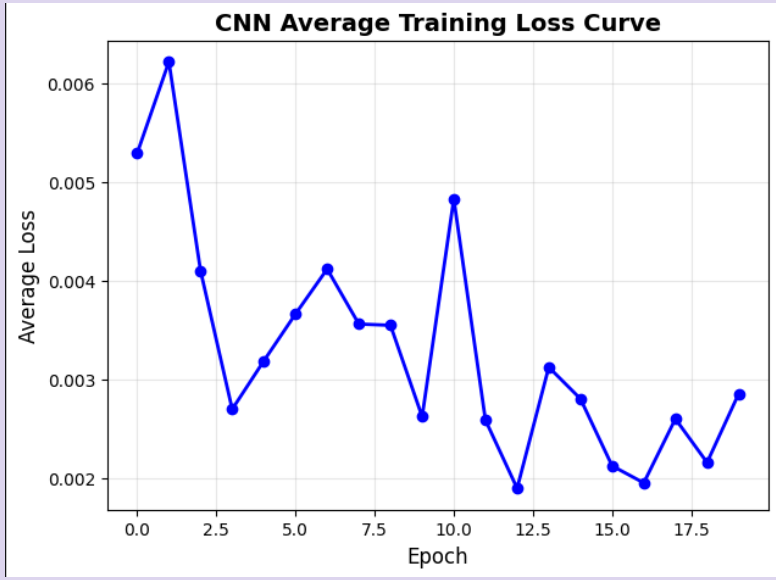
Layer	Input Shape	Output Shape	Param #
Conv1d	(None, 32, 200)	(None, 32, 200)	192
BatchNorm1d	(None, 32, 200)	(None, 32, 200)	64
ReLU	(None, 32, 200)	(None, 32, 200)	0
MaxPool1d	(None, 32, 200)	(None, 32, 100)	0
Conv1d	(None, 32, 100)	(None, 64, 100)	10,304
BatchNorm1d	(None, 64, 100)	(None, 64, 100)	128
ReLU	(None, 64, 100)	(None, 64, 100)	0
MaxPool1d	(None, 64, 100)	(None, 64, 50)	0
Conv1d	(None, 64, 50)	(None, 128, 50)	24,704
ReLU	(None, 128, 50)	(None, 128, 50)	0
MaxPool1d	(None, 128, 50)	(None, 128, 25)	0
Flatten	(None, 128, 25)	(None, 3200)	0
Linear	(None, 3200)	(None, 128)	409,728
ReLU	(None, 128)	(None, 128)	0
Dropout	(None, 128)	(None, 128)	0
Linear	(None, 128)	(None, 2)	258

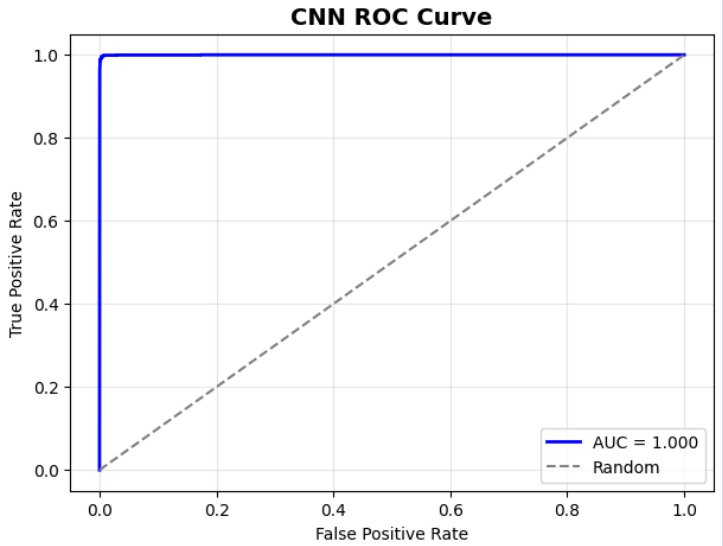
Raw ECG Beat (200 samples) ▶ **Conv1D Blocks (32 → 64 → 128)** ▶ **FC Layer (128 units)** ▶ **Output (Normal / Arrhythmia)**

CNN Model Results & Analysis

99.25%	98.54%	99.80%	1.000
Accuracy	**Precision**	**Recall**	**AUC-ROC**
(4370 / 4403 correct)	*Low false alarm rate*	*Only 4 missed arrhythmias*	*Perfect discrimination*





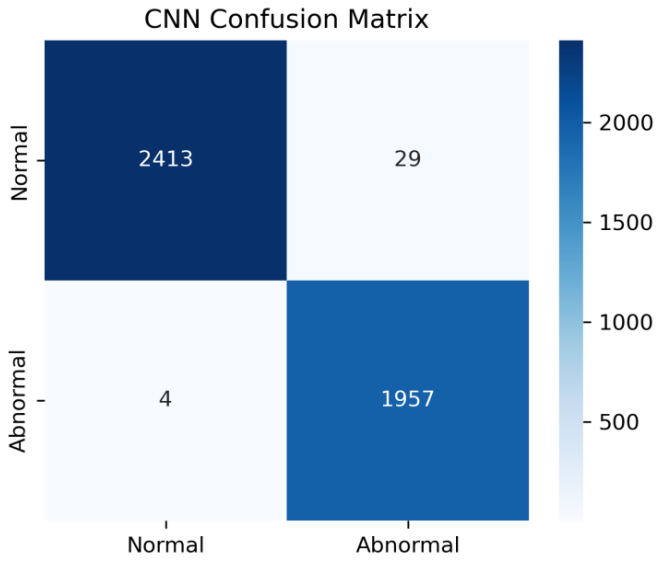


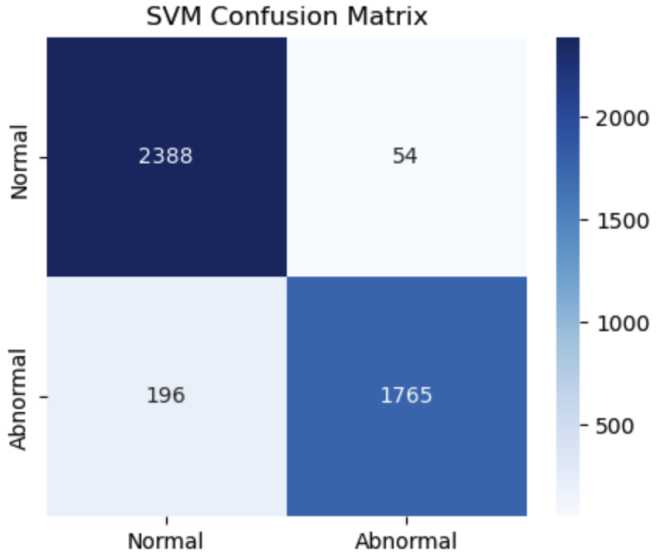
✓ **4 FN** — only 4 arrhythmias missed out of 1961
✓ **29 FP** — 29 false alarms from 2442 normal beats

⚠ **Oscillating loss** — caused by 1:8 class weight & small batch size
✓ **Downward trend** confirms model is converging over 20 epochs

✓ **AUC = 1.000** — perfect separation of classes
✓ **Curve hugs top-left** corner across all decision thresholds

Model Results

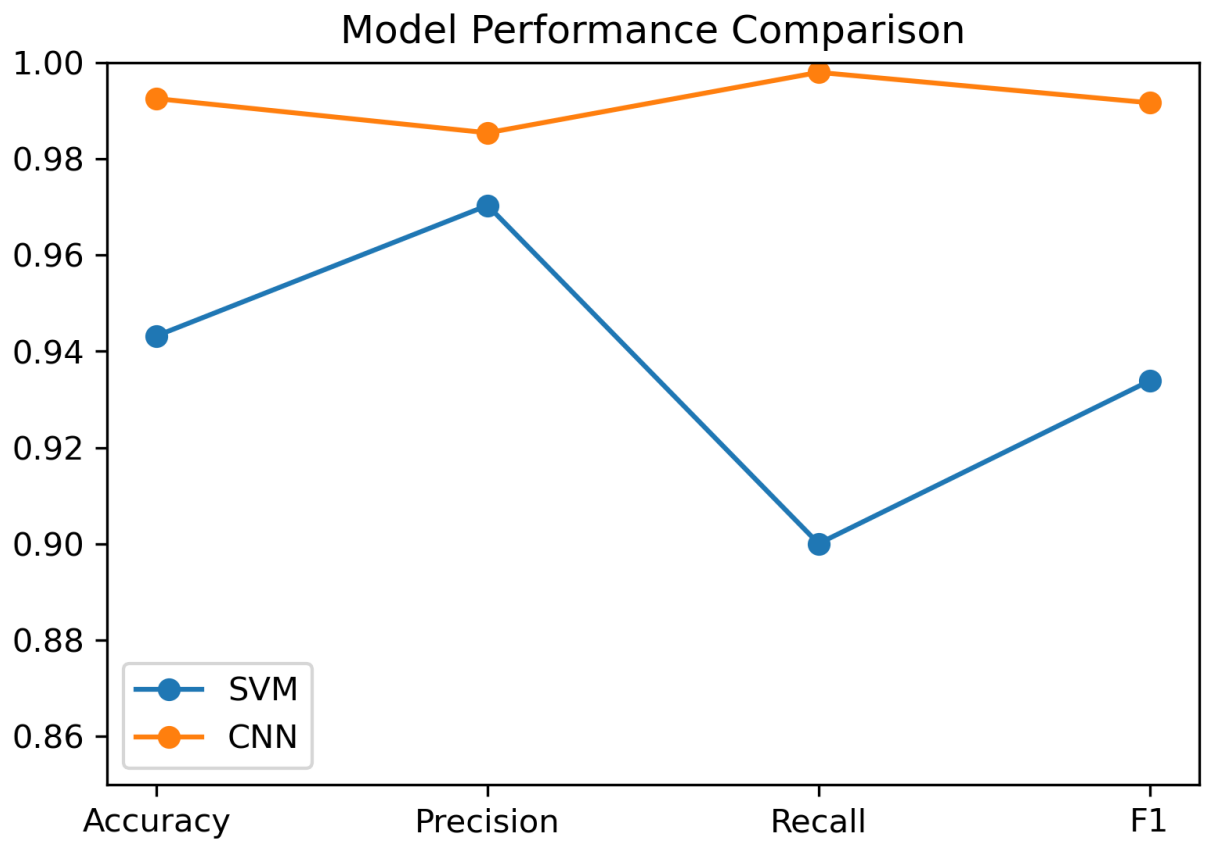




Confusion Matrix Comparison

- **CNN significantly reduces false negatives**
- **SVM misses many abnormal cases**
- **CNN achieves much higher recall**
- **CNN is more suitable for ECG abnormality detection**

Model Results



Model Performance Comparison

- CNN outperforms SVM across all evaluation metrics
- The most significant improvement is observed in recall
- SVM shows lower recall, indicating missed abnormal cases
- CNN is more suitable for ECG abnormality detection

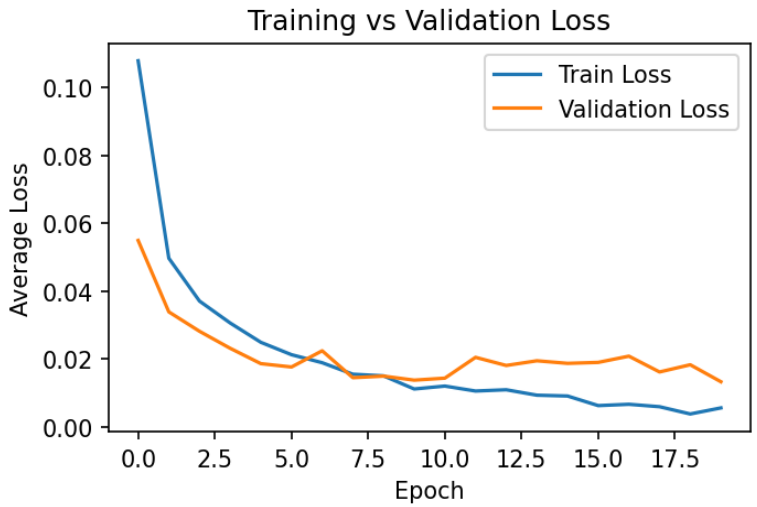
Model Results

Training Loss Curve

- Training loss stabilizes but test loss remains ~2–4× higher with high variance, suggesting the 20% test set is too small for reliable evaluation

- With the 64/16/20 split, both training and validation loss converge smoothly and closely, confirming the model generalizes well without overfitting

- Comparing both splits, the 64/16/20 configuration produces significantly more stable and trustworthy loss curves, making it the stronger evaluation strategy



Conclusion

SVM VS 1D CNN

Best choice for this project: 1D CNN

Due to the importance of correctly diagnosing medical diagnoses and avoiding false negative diagnoses

SVM

- **Simpler pipeline** and easier to explain step by step.
- **Good performance overall**; 97% accuracy is still strong.
- **Balancing improved recall a lot,** from 76% to 94%

1D CNN

- **Best performance** in your comparison across all reported metrics.
- **Very high recall,** which is especially important in medical abnormality detection.
- **Very few false negatives,** so it misses far fewer abnormal beats.

Individual Contributions

Zaid: SVM coding and analysis, CNN coding and analysis, Figure creation, Optimization, Administrative Work

Kaijun: Data preprocessing, CNN implementation and training pipeline, loss optimization, model evaluation and comparison (including confusion matrix), visualization, poster design

Brandon: Evaluation and interpretation of CNN and SVM models, comparison of CNN and SVM models, assisted with poster and slides